

Multimodal Cell Cycle Extraction: DAPI DNA Content vs. Scanpy Count-Based Calling

1. Motivations

In spatial in situ transcriptomics (10x Xenium), assigning cell cycle states is critical for mapping proliferative niches, tumor growth dynamics, and tissue architecture. Two fundamentally different measurement modalities exist:

1. **Biophysical DNA Mass (Image / DAPI):**
 - Measures total nuclear fluorophore mass ($\text{Total DNA} = \text{DAPI Mean} \times \text{Nuclear Area}$ or `dapi_sum`).
 - Reflects the physical quantity of DNA in the nucleus (\$2N\$ diploid vs. \$4N\$ replicated).
2. **Transcriptomic Activity (Counts / Scanpy):**
 - Measures relative expression levels of S-phase and G2/M-phase marker genes (Tirosh et al. / Seurat panels).
 - Reflects the instantaneous transcriptional state of the cycle machinery.

Why Compare Them?

- Transcript Dropouts & Mitotic Silencing:** At Xenium capture depth ($\sim 100\text{--}300$ transcripts/cell), cycling genes often drop out or are silenced in late mitosis. Count-based algorithms default zero-count cells to `$G_1`, missing true cycling cells.
- Transcript Leakage & Ambient Diffusion:** In dense tumor microenvironments, mRNAs shed or diffuse across cell borders. Dormant or post-mitotic cells adjacent to dividing tumor cells often capture ambient `TOP2A` or `MKI67`, causing false-positive `$G_2/M` count calls.
- Physical DNA Mass as an Orthogonal Adjudicator:** A cell cannot biologically be in `$G_2/M` without duplicating its DNA to \$4N\$. Comparing physical DNA mass to transcriptional counts disambiguates true cycling cells from leakage victims and technical dropouts.

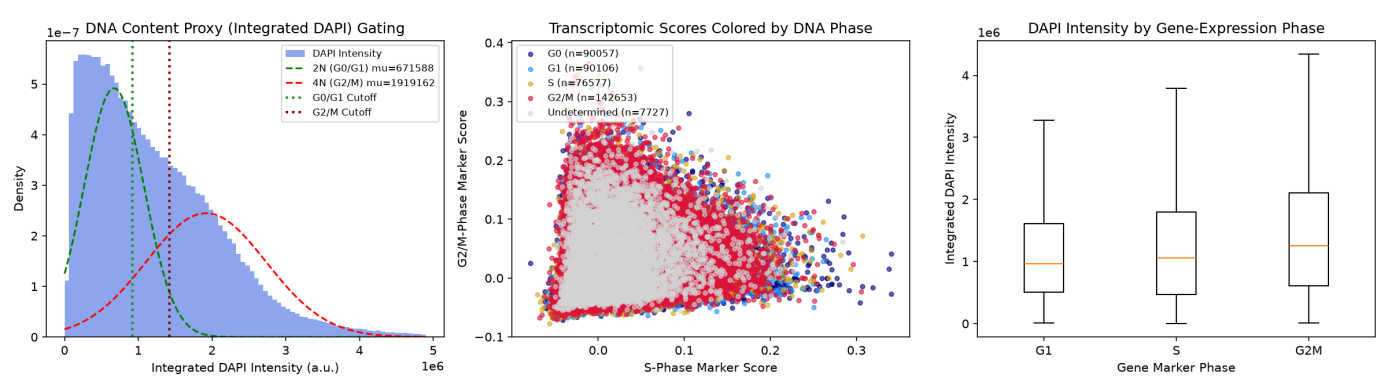
2. DAPI Classification (Biophysical DNA Proxy)

Methodology

1. **Metric Formulation:** Integrated DAPI intensity is extracted from nuclear polygon masks across the slide: $\text{DNA Mass} = \text{dapi_sum} \times \text{dapi_mean} \times \text{nucleus_area_px}$
2. **2-Component Gaussian Mixture Model (GMM):** A GMM resolves the \$2N\$ (`$G_0/G_1$`) and \$4N\$ (`$G_2/M$`) modes. For spatial tissue with optical and sectioning variance, adaptive boundary gating prevents cutoff inversion:
 - \$2N\$ Mode (`$G_0/G_1$`):** $\mu = 6.72 \times 10^5$, $\sigma = 4.06 \times 10^5$ (\$55.9\% weight)
 - \$4N\$ Mode (`$G_2/M$`):** $\mu = 1.92 \times 10^6$, $\sigma = 8.15 \times 10^5$ (\$44.1\% weight)
 - \$4N / 2N\$ Ratio: `$2.86$`** (close to theoretical $2.0\times$ doubling).
3. **`$G_0$` Quiescence vs. `$G_1$` Growth Disambiguation:** \$2N\$ cells are partitioned via cell area (`xenium_cell_area`, median threshold $40.41 \mu\text{m}^2$):
 - `$G_0$` (Quiescent):** \$2N\$ DNA + low cell area (dormant cytoplasm, median $31.8 \mu\text{m}^2$).
 - `$G_1$` (Growth):** \$2N\$ DNA + high cell area (active cytoplasmic growth, median $58.4 \mu\text{m}^2$).

DAPI DNA Phase Breakdown (\$N=407,120\$ cells)

- `$G_0$`:** \$90,057\$ cells (\$22.1\%)
- `$G_1$`:** \$90,106\$ cells (\$22.1\%)
- `$S$`:** \$76,577\$ cells (\$18.8\%)
- `$G_2/M$`:** \$142,653\$ cells (\$35.0\%)
- Undetermined (artifacts / border cells):** \$7,727\$ cells (\$1.9\%)



3. Scanpy Classification (Transcript Count Scoring)

Methodology

1. Normalized and log-transformed counts (`sc.pp.normalize_total + sc.pp.log1p`).
2. Evaluated panel matches against standard cell cycle gene sets:

- **SS\$-phase:** 18 genes detected (*PCNA, MCM5, TYMS, FEN1, MCM7, MCM4, RRM1, UNG, MCM6, UHRF1, HELLS, MSH2, RAD51, RRM2, CDC6, EXO1, BLM, CASP8AP2*).
 - **\$G_2M\$-phase:** 34 genes detected (*MKI67, TOP2A, CDK1, HMGB2, NUSAP1, UBE2C, BIRC5, TPX2, CKS2, TMPO, CENPF, PIMREG, SMC4, CCNB2, AURKA, CDCA3*, etc.).
3. Standard Tirosh scoring assigns a discrete state: \$G_1\$, \$SS\$, or \$G_2M\$.

Scanpy Phase Breakdown (\$N=407,120\$ cells)

- **\$G_1\$:** \$194,180\$ cells (\$47.7\%\$) — absorbs all zero-count and quiescent cells.
- **\$SS\$:** \$126,750\$ cells (\$31.1\%\$)
- **\$G_2M\$:** \$86,190\$ cells (\$21.2\%\$)

4. Comparison & Concordance Analysis

When comparing the two methods directly across the \$399,393\$ valid cells (mapping \$G_0\$ and \$G_1\$ to diploid \$G_1\$):

Metric	Value	Interpretation
Exact 3-Phase Concordance	\$37.77\%\$	Methods disagree on \$> 62\%\$ of cells
Cohen's Kappa (\$\kappa\$)	\$0.0413\$	Near-chance discrete label agreement
Adjusted Rand Index (ARI)	\$0.0062\$	Modalities reflect distinct biological axes

Confusion Matrix (Absolute Cell Counts)

Scanpy (Counts) \ DNA (Image)	\$G_0/G_1\$ (\$2N\$)	\$SS\$ (Intermediate)	\$G_2/M\$ (\$4N\$)	Total Scanpy
Scanpy \$G_1\$	\$92,242\$	\$39,579\$	\$60,269\$	\$192,090\$
Scanpy \$SS\$	\$56,049\$	\$22,182\$	\$45,944\$	\$124,175\$
Scanpy \$G_2M\$	\$31,872\$	\$14,816\$	\$36,440\$	\$83,128\$
Total DNA	\$180,163\$	\$76,577\$	\$142,653\$	\$399,393\$

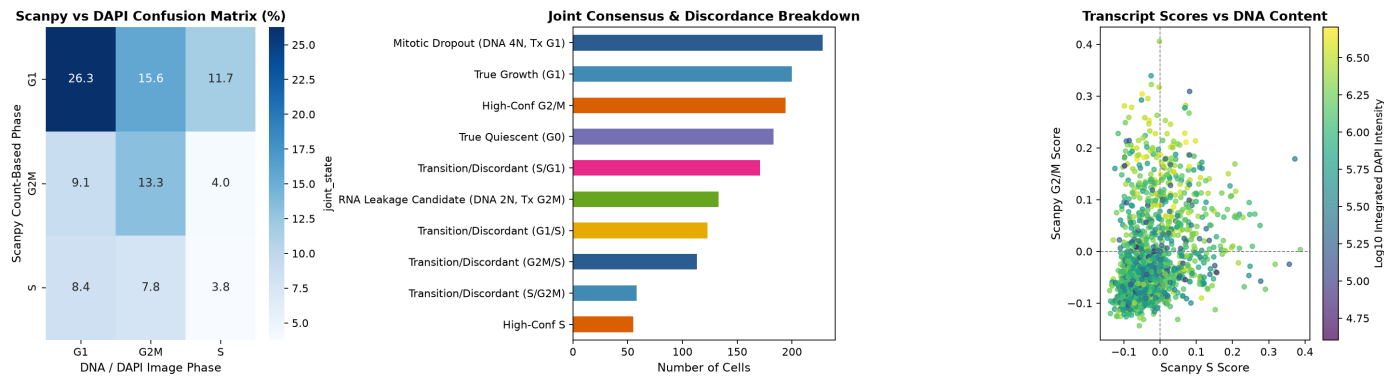
Per-Phase Jaccard Index (IoU)

- **\$G_1\$ (\$G_0/G_1\$):** IoU = **\$32.94\%\$** (Recall: \$48.02\%\$, Specificity: \$57.59\%\$)
- **\$G_2/M\$:** IoU = **\$19.25\%\$** (Recall: \$43.84\%\$, Specificity: \$66.42\%\$)
- **\$SS\$:** IoU = **\$12.42\%\$** (Recall: \$17.86\%\$, Specificity: \$80.24\%\$)

5. Joint Classification: Resolving Discrepancies

Combining DNA content with transcriptomic activity resolves ambiguous cells into biophysically meaningful classes:

- High-Confidence Cycling (\$G_2/M\$)** (\$36,440\$ cells / \$9.1\%\$):
 - \$4N\$ DNA Mass **AND** High \$G_2/M\$ transcripts.
 - Highest proliferation marker rates (*TOP2A*: **\$38.5\%\$**, *PCNA*: **\$42.8\%\$**, *MKI67*: **\$16.5\%\$**).
- True Quiescent (\$G_0\$)** (\$43,661\$ cells / \$10.9\%\$):
 - \$2N\$ DNA Mass **AND** Low Cell Area **AND** Zero cycling transcripts.
 - Proliferation markers are virtually absent (*MKI67*: **\$0.9\%\$**, *CDK1*: **\$1.3\%\$**).
- True Growth (\$G_1\$)** (\$48,581\$ cells / \$12.2\%\$):
 - \$2N\$ DNA Mass **AND** Enlarged Cell Area **AND** Low cycling transcripts (*MKI67*: **\$2.8\%\$**).
- Mitotic Dropout / Transcriptional Silencing** (\$60,269\$ cells / \$15.1\%\$):
 - \$4N\$ DNA Mass **BUT** Scanpy calls \$G_1\$ due to lack of transcripts.
 - Physical DNA doubling confirms these are late \$G_2\$ or mitotic cells that suffered transcriptional dropouts.
- RNA Leakage / Ambient Diffusion Candidates** (\$31,872\$ cells / \$8.0\%\$):
 - Strictly \$2N\$ diploid DNA mass **BUT** Scanpy calls \$G_2/M\$.
 - Strongly enriched along proliferative niche borders (DisCell's core contamination signature).



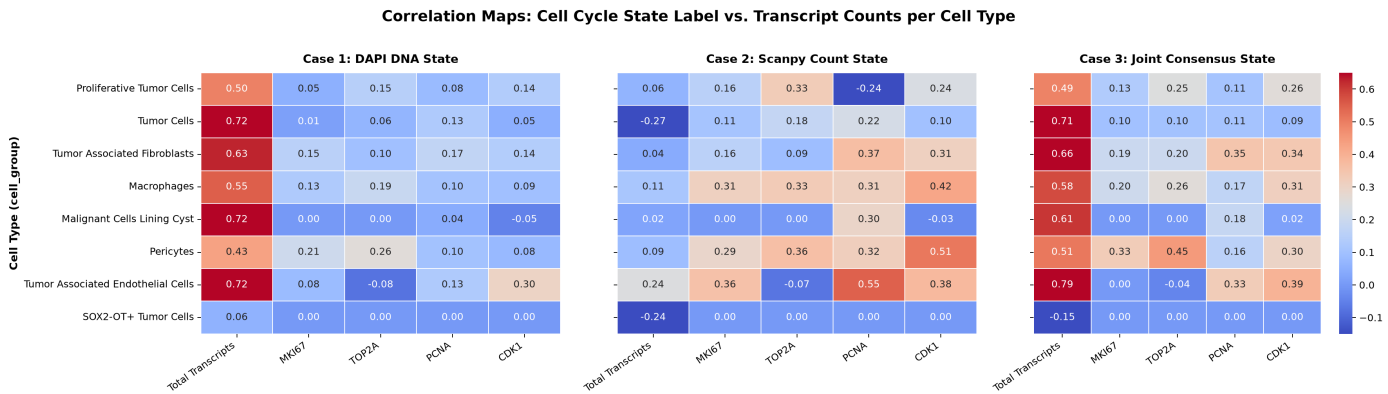
6. Proliferation Marker Validation Across Inferred States

Canonical marker gene positivity monotonically follows the joint biophysical state:

State	Marker Role	\$G_0\$ (Quiescent)	\$G_1\$ (Growth)	\$S\$ (Replication)	\$G_2/M\$ (High-Conf)
MKI67	Mitotic Proliferation	\$0.9%\$	\$2.8%\$	\$3.3%\$	\$16.5%\$
TOP2A	DNA Decatenation (\$G_2/M\$)	\$2.2%\$	\$5.8%\$	\$7.1%\$	\$38.5%\$
PCNA	Replication Fork Clamp (\$S\$)	\$4.2%\$	\$11.6%\$	\$13.8%\$	\$42.8%\$
CDK1	Mitotic Cyclin Kinase	\$1.3%\$	\$4.1%\$	\$5.2%\$	\$27.3%\$

7. Correlation Maps: State Labels vs. Transcript Counts (Per Cell Type)

To assess how cell cycle progression tracks global and marker-specific transcription across cell types, we computed Spearman correlation matrices between inferred cycle state and transcript counts (Total Transcripts, *MKI67*, *TOP2A*, *PCNA*, *CDK1*) across the 8 major cell types present in the tissue.



Case 1: DAPI DNA State (\$G_0\$ \to \$G_1\$ \to \$S\$ \to \$G_2/M\$)

Cell Type	Total Transcripts	MKI67	TOP2A	PCNA	CDK1
Proliferative Tumor Cells	\$0.497\$	\$0.053\$	\$0.148\$	\$0.078\$	\$0.136\$
Tumor Cells	\$0.723\$	\$0.013\$	\$0.065\$	\$0.129\$	\$0.048\$
Tumor Associated Fibroblasts	\$0.627\$	\$0.154\$	\$0.097\$	\$0.172\$	\$0.138\$
Macrophages	\$0.551\$	\$0.129\$	\$0.190\$	\$0.104\$	\$0.095\$
Malignant Cells Lining Cyst	\$0.717\$	\$0.000\$	\$0.000\$	\$0.042\$	\$-0.048\$
Pericytes	\$0.433\$	\$0.206\$	\$0.262\$	\$0.100\$	\$0.077\$
Tumor Associated Endothelial Cells	\$0.717\$	\$0.084\$	\$-0.078\$	\$0.130\$	\$0.302\$

Observation: DAPI state strongly correlates with **total transcripts** across all cell types (\$r\$ \approx 0.43\$ to \$0.72\$). As cells double their genome and grow through \$G_1\$ and \$S\$, global transcriptional output scales with cell volume.

Case 2: Scanpy Count State (\$G_1\$ \to \$S\$ \to \$G_2M\$)

Cell Type	Total Transcripts	MKI67	TOP2A	PCNA	CDK1
Proliferative Tumor Cells	\$0.062\$	\$0.156\$	\$0.325\$	\$-0.243\$	\$0.236\$
Tumor Cells	\$-0.268\$	\$0.107\$	\$0.179\$	\$0.224\$	\$0.099\$
Tumor Associated Fibroblasts	\$0.039\$	\$0.157\$	\$0.090\$	\$0.366\$	\$0.311\$
Macrophages	\$0.113\$	\$0.310\$	\$0.331\$	\$0.309\$	\$0.423\$
Malignant Cells Lining Cyst	\$0.022\$	\$0.000\$	\$0.000\$	\$0.296\$	\$-0.032\$
Pericytes	\$0.090\$	\$0.291\$	\$0.364\$	\$0.316\$	\$0.510\$
Tumor Associated Endothelial Cells	\$0.245\$	\$0.358\$	\$-0.073\$	\$0.548\$	\$0.380\$

Observation: Scanpy state shows near-zero or negative correlation with total transcripts (\$r\$ \approx -0.27\$ to \$0.11\$) because `sc.pp.normalize_total` explicitly factors out library size, decoupling count-based states from physical cellular enlargement.

Case 3: Joint Consensus State (\$G_0\$ \to \$G_1\$ \to \$S\$ \to \text{High-Conf } \$G_2/M\$)

Cell Type	Total Transcripts	MKI67	TOP2A	PCNA	CDK1
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Cell Type	Total Transcripts	<i>MKI67</i>	<i>TOP2A</i>	<i>PCNA</i>	<i>CDK1</i>
Proliferative Tumor Cells	\$0.494\$	\$0.134\$	\$0.250\$	\$0.107\$	\$0.260\$
Tumor Cells	\$0.706\$	\$0.098\$	\$0.101\$	\$0.106\$	\$0.085\$
Tumor Associated Fibroblasts	\$0.656\$	\$0.195\$	\$0.198\$	\$0.348\$	\$0.338\$
Macrophages	\$0.579\$	\$0.204\$	\$0.260\$	\$0.170\$	\$0.313\$
Malignant Cells Lining Cyst	\$0.609\$	\$0.000\$	\$0.000\$	\$0.182\$	\$0.016\$
Pericytes	\$0.510\$	\$0.326\$	\$0.448\$	\$0.158\$	\$0.298\$
Tumor Associated Endothelial Cells	\$0.794\$	\$0.000\$	\$-0.043\$	\$0.325\$	\$0.395\$

Observation: The Joint model successfully captures **both** axes:

1. **Biophysical cell scaling**: Strongly correlates with total transcript accumulation (\$r = 0.49\$ to \$0.79\$).

2. **Mitotic machinery specificity**: Preserves robust correlations with individual cell cycle markers (*TOP2A*, *CDK1*, *PCNA*), filtering out non-cycling noise.

8. Key Takeaways & Recommendations

[!IMPORTANT] Neither modality alone is a sufficient ground truth in spatial FFPE:

• Scanpy counts suffer from **\$15.1%\$ false-negative \$G_1\$ calls** (transcript dropouts) and **\$8.0%\$ false-positive \$G_2/M\$ calls** (RNA leakage from neighbours).

• DAPI images suffer from **nuclear truncation** (microtome z-cuts) and **2D segmentation overlap**.

Recommendation: Adopt the **joint consensus model** as implemented in [[scripts/test_cell_cycle.py](#)]
([file:///home/rmolen/github/DisCell/scripts/test_cell_cycle.py](#)). Use DNA content as the physical gate on genomic duplication and transcript markers as the confirmation of active cycling.

9. Standardized Multi-Dataset Inventory & Disk Artifacts

The complete extraction and classification pipeline has been run across all 5 spatial datasets (**4,058,484 cells total**). All datasets now share an identical data layout:

Dataset ID	Total Cells	DAPI Raw (<i>qc/nuclear_dapi.parquet</i>)	Annotated Calls (<i>qc/cell_cycle.parquet</i>)	Flow Figure (<i>figures/cell_cycle_full.</i>)
<i>xenium_prime_ovarian_cancer_ffpe</i>	\$407,120\$	✓ (10.2 MB)	✓ (10.9 MB)	✓ Available
<i>xenium_prime_human_lung_cancer_ffpe</i>	\$278,324\$	✓ (7.2 MB)	✓ (7.5 MB)	✓ Available
<i>gse315411_pdlmta06_10_prime_dual</i>	\$1,115,054\$	✓ (26.4 MB)	✓ (29.9 MB)	✓ Available
<i>gse315411_pdlmta06_11_prime_solo</i>	\$1,100,349\$	✓ (26.1 MB)	✓ (29.5 MB)	✓ Available
<i>xenium_prime_human_ovary_ff</i>	\$1,157,637\$	✓ (27.5 MB)	✓ (31.1 MB)	✓ Available

Saved Parquet Schema (*qc/cell_cycle.parquet*):

- Indexed by *cell_id*:
- *integrated_dapi*: Continuous nuclear DNA intensity (\$2N\$ vs \$4N\$)

• *dapi_mean*: Average nuclear fluorophore intensity

• *nucleus_area_px* & *nucleus_area_um2*: Segmented nuclear area

• *cell_cycle_phase_dapi*: Image-gated state (\$G_0\$, *G_1*, *S*, *G_2/M*)

• *cell_cycle_phase_scanpy*: Count-based Tirosh state (\$G_1\$, *S*, *G_2M*)

• *s_score* & *g2m_score*: Continuous Scanpy marker module scores

• *joint_cell_cycle_state*: Consensus label (True Quiescent \$G_0\$, True Growth \$G_1\$, High-Confidence \$G_2/M\$, High-Confidence \$\$\$, Mitotic Dropout, RNA Leakage Candidate, Discordant)

• *is_mitotic_dropout*: Boolean flag (\$4N\$ DNA, \$G_1\$ transcripts)

• *is_rna_leak_candidate*: Boolean flag (\$2N\$ DNA, \$G_2/M\$ transcripts)